

What is machine learn



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What is machine learning? Machine learning



What is machine lea

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Machine learning (ML) is a branch of computer science that focuses on enabling computers and machines to learn, to perform tasks autonomously and accurately through experience and data.

[UC Berkeley](#) breaks out the learning system into three main parts.

1. A Decision Process: In general, machine learning is used for prediction or classification. Based on some unlabeled data, your algorithm will produce an output.
2. An Error Function: An error function evaluates the model's performance on known examples, an error function can measure the model's performance.
3. A Model Optimization Process: If the model's performance is not good on the training set, then weights are adjusted to improve the model's performance. The algorithm "optimize" process, updating weights automatically until the performance has been met.

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Machine learning ve versus neural netwc

Since [deep learning](#) and machine learning ter noting the nuances between the two. Machine [networks](#) are all sub-fields of artificial intellig sub-field of machine learning, and deep learn

The way in which deep learning and machine learns. "Deep" machine learning can use labe learning, to inform its algorithm, but it doesn' deep learning process can ingest unstructure and it can automatically determine the set of categories of data from one another. This elin required and enables the use of large amount "scalable machine learning" as Lex Fridman n

Classical, or "non-deep," machine learning is learn. Human experts determine the set of fe: between data inputs, usually requiring more s

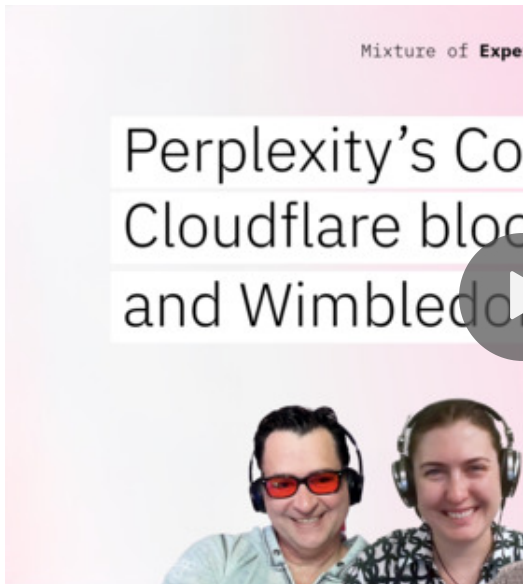
Neural networks, or artificial neural networks containing an input layer, one or more hidden artificial neuron, connects to another and has output of any individual node is above the spe activated, sending data to the next layer of th along to the next layer of the network by that referring to the number of layers in a neural n more than three layers, which would be inclu: considered a deep learning algorithm or a dee only has three layers is just a basic neural net

Deep learning and neural networks are credit as [computer vision](#), [natural language process](#)

See the blog post "[AI vs. Machine Learning vs the Difference?](#)" for a closer look at how the c

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Machine learning m

Machine learning models fall into three prima

Supervised learning

Supervised learning, also known as supervise labeled datasets to train algorithms to classif input data is fed into the model, the model ac appropriately. This occurs as part of the cross avoids **overfitting** or **underfitting**. Supervised of real-world problems at scale, such as class

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inbox. Some methods used in supervised learning are [linear regression](#), [logistic regression](#), [random forest](#), and [support vector machines](#).

Unsupervised learning

[Unsupervised learning](#), also known as unsupervised learning algorithms to analyze and cluster unlabeled data. These algorithms discover hidden patterns or associations in the data without the need for human intervention.

Unsupervised learning's ability to discover hidden patterns make it ideal for exploratory data analysis, customer segmentation, and image and pattern recognition. It finds features in a model through the process of dimensionality reduction ([PCA](#)) and singular value decomposition. Other algorithms used in unsupervised learning include [k-means clustering](#), and probabilistic clustering methods like [Gaussian mixture models](#).

Semi-supervised learning

Semi-supervised learning offers a happy medium between supervised and unsupervised learning. During training, it uses a smaller labeled dataset for feature extraction from a larger, unlabeled dataset. This helps overcome the problem of not having enough labeled data, which is often the case in real-world applications where labeling data is too costly.

For a deep dive into the differences between supervised and unsupervised learning, read [Unsupervised Learning: What's the Difference?](#)

Reinforcement learning

Reinforcement learning is a machine learning paradigm where the algorithm isn't trained using sample data but learns through trial and error. A sequence of successful outcomes leads to a reward, which is used to recommend a policy for a given problem.

The IBM Watson® system that won the *Jeopardy!* tournament used reinforcement learning to learn from its mistakes and improve its performance.

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as it were), which square to select on the board daily doubles.

Common machine learning

A number of machine learning algorithms are

1. Neural networks
2. Linear regression
3. Logistic regression
4. Clustering
5. Decision trees
6. Random forests

Neural networks

Neural networks simulate the way the human processing nodes. Neural networks are good at an important role in applications including natural speech recognition, and image creation.

Linear regression

This algorithm is used to predict numerical values at different values. For example, the technique can be used on historical data for the area.

Logistic regression

This supervised learning algorithm makes **predictions** such as “yes/no” answers to questions. It can be used for spam and quality control on a production line.

Clustering

Using unsupervised learning, clustering algorithms can be grouped. Computers can help data scientists

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data items that humans have overlooked.

Decision trees

Decision trees can be used for both predicting and classifying data into categories. Decision tree decisions that can be represented with a tree structure is that they are easy to validate and audit.

Random forests

In a random forest, the machine learning algorithm combines the results from a number of decision trees.

Advantages and disadvantages of machine learning algorithms

Depending on your budget, need for speed and accuracy, you can choose between supervised, unsupervised, semi-supervised, and reinforcement learning algorithms.

For example, decision tree algorithms are used for regression problems and classifying data into categories. A sequence of linked decisions that may be represented by a tree structure. One advantage of decision trees is that they are easy to validate and audit. The bad news is that they can be overfitted.

Overall, there are many advantages to machine learning. These include machine learning algorithms that can process massive volumes of data that humans might not be able to handle. Just feed in the dataset and let the system assemble and refine its own algorithm based on the data input over time. Customers and users can interact with the model, and the model learns more with every experience.

On the downside, machine learning requires a lot of data and computing power.

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unbiased. GIGO is the operative factor: garbage in, garbage out, and having a system robust enough to run it n

Machine learning can also be prone to error, c
sample, the system could produce a perfectly
or misleading. To avoid wasting budget or dis
on the answers only when there is high confic

Real-world machine

Here are just a few examples of machine lear

Generative AI: [Generative AI](#), or gen AI, is m
content—text, images, video, software code—
Gen AI relies on deep learning models that id
relationships in huge amounts of data, and th
requests and create new content. [ChatGPT](#) ar
apps.

AI agents and agentic AI: An [AI agent](#) is an a
tasks and accomplish goals on behalf of a use
intervention, by designing its own workflow a
or services). [Agentic AI](#) is a system of multipl
coordinated, or orchestrated, to accomplish a
any single agent in the system could accompl

[Explore our 2025 guide to AI agents](#) →

Speech recognition: Also known as automati
speech recognition, or speech-to-text, [speech](#)
[processing](#) (NLP) to translate human speech i
incorporate speech recognition into their syst
improve accessibility for texting.

Customer service: [Online chatbots](#) are replac
journey, changing the way we think about cus
social media platforms. Chatbots answer freq
such as shipping, or provide personalized adv
sizes for users. Examples include [virtual agen](#)

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using Slack and Facebook Messenger; and tax voice assistants.

Computer vision: This [AI](#) technology enables from digital images, videos, and other visual i Powered by convolutional neural networks, [c](#) tagging on social media, radiology imaging in automotive industry.

Recommendation engines: Using past consu to discover data trends that can be used to de strategies. Recommendation engines are use product recommendations to customers durin

Robotic process automation (RPA): Also kno automation technologies to perform repetitiv

Automated stock trading: Designed to optim frequency trading platforms make thousands human intervention.

Fraud detection: Banks and other financial ins suspicious transactions. Supervised learning known fraudulent transactions. Anomaly dete atypical and deserve further investigation.

Challenges of mach

As machine learning technology has develop However, implementing machine learning in t ethical concerns about AI technologies. Some

Technological singularity

While this topic garners a lot of public attentio the idea of AI surpassing human intelligence i also referred to as strong AI or superintelligence superintelligence as “any intellect that vastly practically every field, including scientific cre:

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Despite the fact that superintelligence is not i
some interesting questions as we consider th
driving cars. It's unrealistic to think that a dri
who is responsible and liable under those circ
autonomous vehicles, or do we limit this tech
help people drive safely? The jury is still out o
debates that are occurring as new, innovative

AI impact on jobs

While a lot of public perception of artificial int
concern should probably be reframed. With e
the market demand for specific job roles shift
automotive industry, many manufacturers, lik
production to align with green initiatives. The
source of energy is shifting from a fuel econom

In a similar way, artificial intelligence will shif
will need to be individuals to help manage AI
address more complex problems within the ir
by job demand shifts, such as customer servi
intelligence and its effect on the job market w
roles that are in demand.

Privacy

Privacy tends to be discussed in the context c
security. These concerns have allowed policy
For example, in 2016, GDPR legislation was c
in the European Union and European Econom
their data. In the United States, individual sta
California Consumer Privacy Act (CCPA), whic
businesses to inform consumers about the cc
has forced companies to rethink how they sto
information (PII). As a result, investments in :
for businesses as they seek to eliminate any \
surveillance, hacking, and cyberattacks.

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Bias and discrimination

Instances of bias and discrimination across a raised many ethical questions regarding the u safeguard against bias and discrimination wh by biased human processes? While companie automation efforts, [Reuters](#) ² highlights so incorporating AI into hiring practices. In their Amazon unintentionally discriminated agains roles, and the company ultimately had to scre [Review](#) ³ has raised other pointed question as what data you should be able to use when

Bias and discrimination aren't limited to the h found in a number of applications from facial algorithms. As businesses become more awa more active in this discussion around AI ethic

Accountability

Since there isn't significant legislation to regu enforcement mechanism to ensure that ethic companies to be ethical are the negative repe bottom line. To fill the gap, ethical framework between ethicists and researchers to govern t models within society. However, at the mome Some [research](#) ⁴ shows that the combinati foresight into potential consequences aren't c

How to choose the r machine learning

Selecting a platform can be a challenging pro or limit the use of other valuable tools or tech to select an AI platform, there is often a tende system. Maybe so, but reviewers should start be doing for their organization. What machine and what features are important to accomplis the usefulness of an entire system. Here are s

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MLOps capabilities. Does the system have:

- a **unified interface** for ease of management
- automated machine learning tools for fast code functionality?
- decision optimization to streamline the selection of models?
- visual modeling to combine visual data science with notebook-based interfaces on a unified dashboard
- automated development for beginners to professional scientists to experiment?
- synthetic data generator as an alternative when real-world data is not readily available?

Generative AI capabilities. Does the system have:

- a **content generator** that can generate text, images, or audio data it was trained on?
- automated classification to read and classify documents, sorting customer complaints or reviewing support tickets
- a summary generator that can transform complex data into concise capture key points from financial reports, legal documents, or research papers
- a data extraction capability to sort through large volumes of necessary information from large documents

Report

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Learn why IBM has been recognized as a Leader in the 2025 Gartner® Magic Quadrant™

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for Data Science and Machine Learning Platforms.

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Resources

Report

The 2025 CEO's guide: 5 mindshifts to supercharge business growth

Activate these five mindshifts to cut through the uncertainty, spur business reinvention, and supercharge growth with agentic AI.

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Training

Level up your ML expertise

Learn fundamental concepts and build your skills with hands-on labs,

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Ebook

Unlock the power of generative AI + ML

Learn how to confidently incorporate generative AI and machine learning into your business.

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Guide

Put AI to work: Driving ROI with gen AI

Want to get a better return on your AI investments? Learn how scaling gen AI in key areas drives change by helping your best minds build and deliver innovative new solutions.

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Ebook

How to choose the right foundation model

Learn how to select the most suitable AI foundation model for

AI models

Explore IBM Granite

IBM® Granite™ is our family of open, performant and trusted AI models, tailored for business and optimized to scale your AI applications. Explore language, code, time series and guardrail options.

[Meet Granite](#)



Guide

How to thrive in this new era of AI with trust and confidence

Dive into the 3 critical elements of a strong AI strategy: creating a competitive edge, scaling AI across

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the business and advancing trustworthy AI.

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Report

AI in Action Report

We surveyed 2,000 organizations about their AI initiatives to discover what's working, what's not and how you can get ahead.

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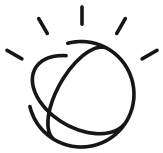


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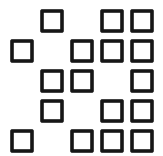
Related solutions

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**IBM watsonx.ai**

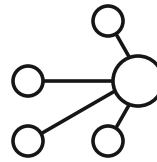
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